

High-Speed Link Design

Lab Report 01

S-Parameters, Group Delay, and 8 Gb/s Pulse Response of a High-Speed Channel

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Cadence Virtuoso Simulation Analysis

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1 Introduction

This lab provides a comprehensive frequency- and time-domain characterization of a multi-port high-speed channel (`channel.s4p`) using Cadence Virtuoso. Part 1 analyzes differential insertion loss (S_{21}/SDD_{21}), wrapped/unwrapped phase behavior, and the extracted channel group delay (τ_g) up to 20 GHz. Part 2 models the transient pulse response at 8 Gb/s (UI = 125 ps) to quantify inter-symbol interference (ISI) and extract signal cursors required for transceiver equalization.

2 Part 1: S-Parameter Analysis

2.1 Methodology

2.1.1 Method 1: Balun-Based Differential Conversion

Two $50\ \Omega$ single-ended ports (PORT1 and PORT2) are coupled to ideal balun blocks (`rflib/balun`) to interface with differential input ports (1 & 3) and output ports (2 & 4) of the `n4port` channel model. An S-parameter sweep is performed from 0 Hz to 20 GHz.

2.1.2 Method 2: Direct 4-Port Mixed-Mode Conversion

Four independent $50\ \Omega$ ports (PORT1–PORT4) drive the `n4port` block directly with reference pins grounded. Mixed-mode matrix conversion computes the differential transmission coefficient:

$$SDD_{21} = \frac{1}{2}(S_{21} - S_{23} - S_{41} + S_{43}) \quad (1)$$

2.2 Results and Analysis

2.2.1 Insertion Loss and Phase Response

Figure 1 illustrates the Method 1 schematic, while Figure 2 shows the differential insertion loss magnitude, reaching approximately $-43.3\ \text{dB}$ at 15 GHz.

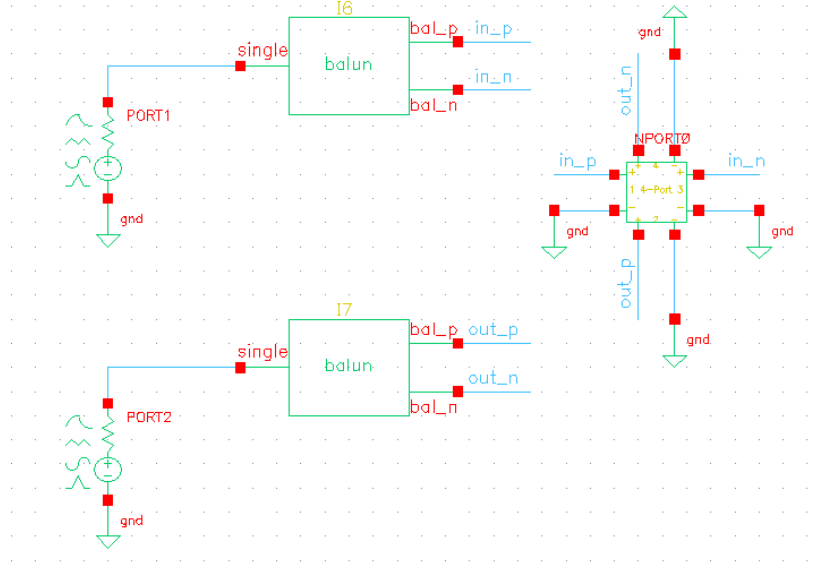


Figure 1: Method 1 schematic configuration using differential baluns.

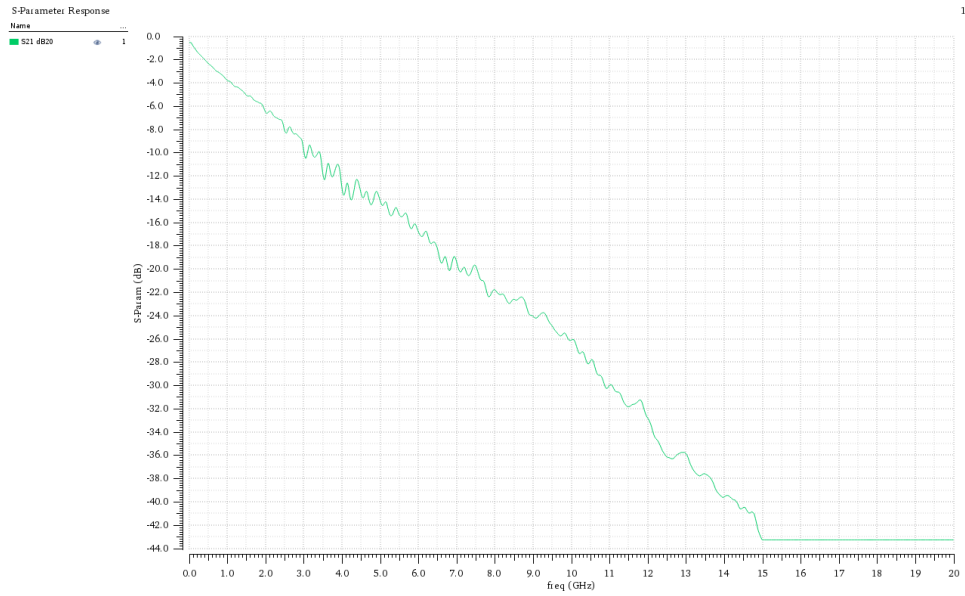


Figure 2: Method 1: Differential insertion loss (S_{21}) in dB20 up to 20 GHz.

The phase response is analyzed in wrapped (Figure 3) and unwrapped forms (Figure 4). Phase wrapping restricts values within $[-180^\circ, +180^\circ]$, introducing artificial 360° phase discontinuities. Unwrapping resolves these jumps, revealing a linear phase accumulation reaching -21.7 kdeg ($\approx -21,700^\circ$) near 15–20 GHz.

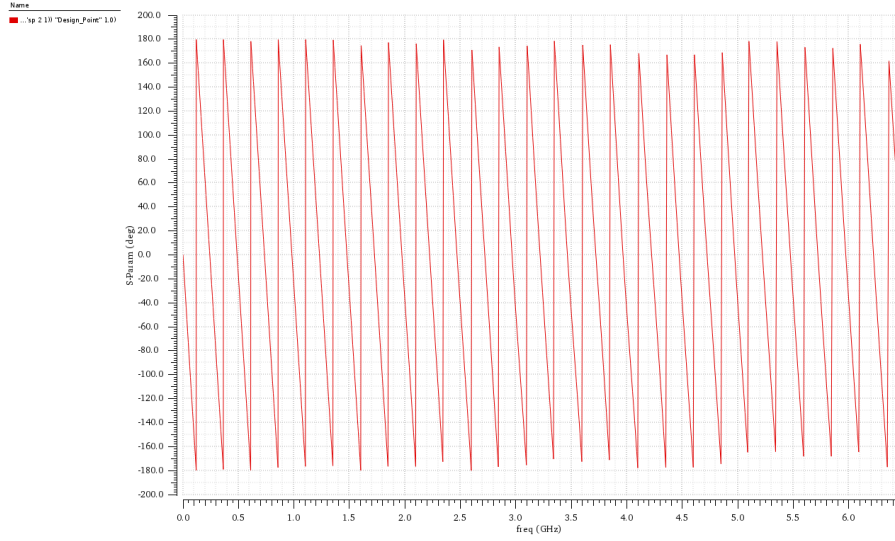


Figure 3: Method 1: Wrapped phase response of S_{21} ($[-180^\circ, +180^\circ]$).

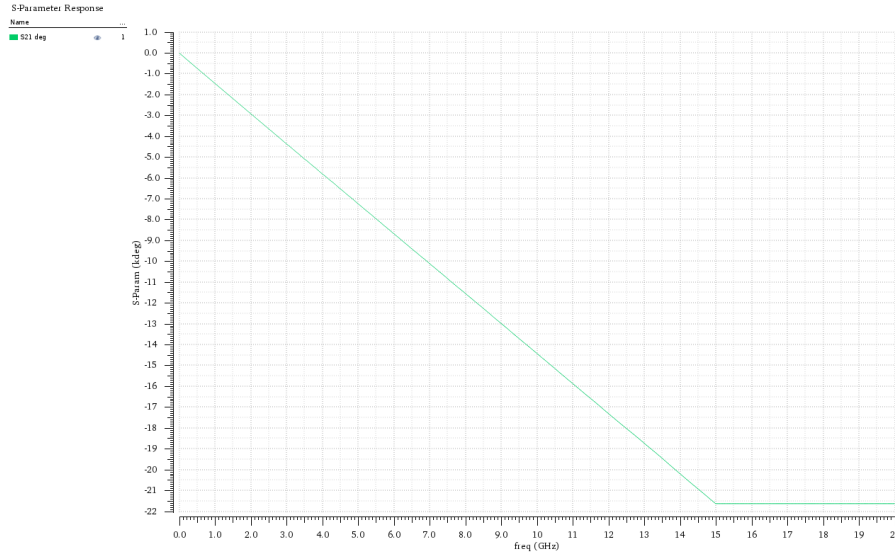


Figure 4: Method 1: Unwrapped phase response showing continuous phase roll-off.

2.2.2 Group Delay Analysis

Group delay (τ_g) quantifies the time delay experienced by individual frequency components as they propagate through the transmission channel. Mathematically, it is defined as the negative derivative of the phase response with respect to angular frequency:

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega} \quad (2)$$

When frequency f is expressed in Hertz (Hz) and phase ϕ_{deg} is measured in degrees, the group delay calculation simplifies to:

$$\tau_g(f) = -\frac{1}{360^\circ} \frac{d\phi_{\text{deg}}(f)}{df} \quad (3)$$

Using the Cadence Virtuoso Calculator, the derivative of the unwrapped S_{21} phase response was evaluated and averaged over the linear operating region from 0 to 15 GHz (Figures 5 and 6). The resulting mean phase slope was found to be:

$$\left\langle \frac{d\phi_{\text{deg}}}{df} \right\rangle \approx -1.441 \times 10^{-6} \frac{\text{deg}}{\text{Hz}} \quad (4)$$

Substituting this mean slope into the group delay relation yields:

$$\tau_g = -\frac{-1.441 \times 10^{-6} \text{ deg/Hz}}{360^\circ/\text{cycle}} \approx 4.0028 \times 10^{-9} \text{ s} \implies \boxed{\tau_g \approx 4.00 \text{ ns}} \quad (5)$$



Figure 5: Unwrapped S_{21} phase and its derivative evaluated across frequency.

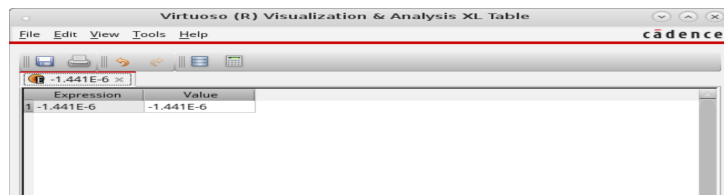


Figure 6: Cadence Virtuoso table showing the average phase derivative ($-1.441 \times 10^{-6} \text{ deg/Hz}$).

This calculated group delay exhibits strong consistency with the time-domain transient pulse response in Part 2. Given that the input excitation pulse starts at $t = 5.0 \text{ ns}$ and the primary differential output peak occurs at $t_{\text{main}} = 9.1043 \text{ ns}$, the physical propagation delay through the channel is approximately $t_{\text{prop}} \approx 4.10 \text{ ns}$.

2.2.3 Method 2 Results

Figure 7 shows the 4-port setup, while Figure 8 plots the SDD_{21} response. Results match Method 1 identically across the frequency band.

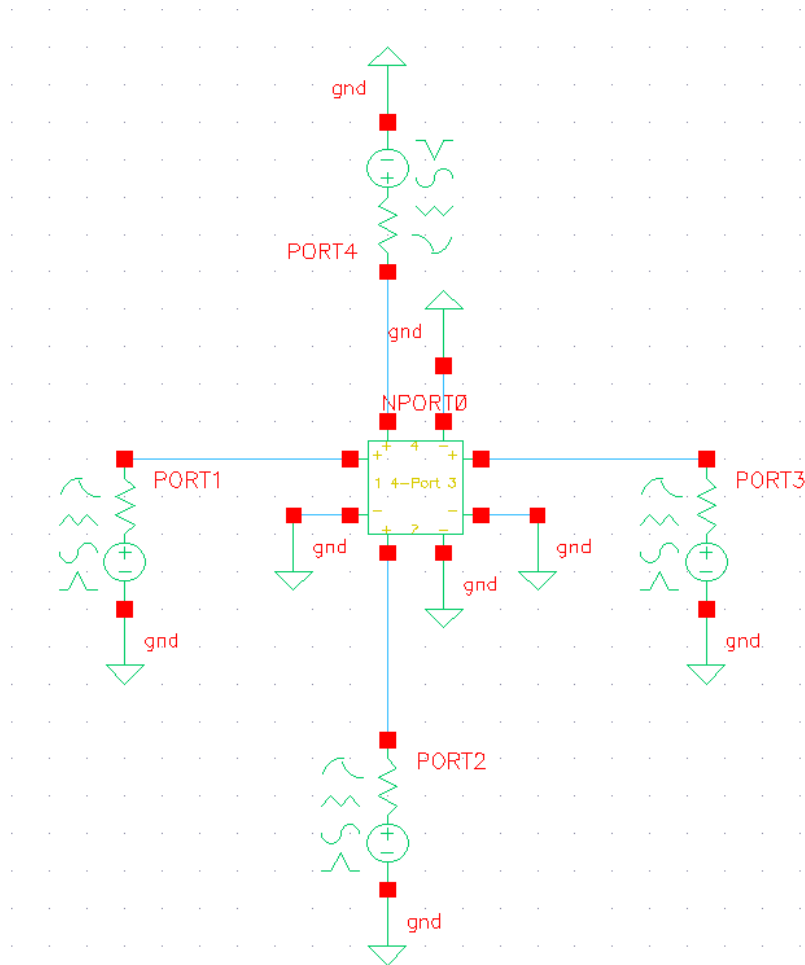


Figure 7: Method 2 schematic with direct 4-port single-ended excitation.

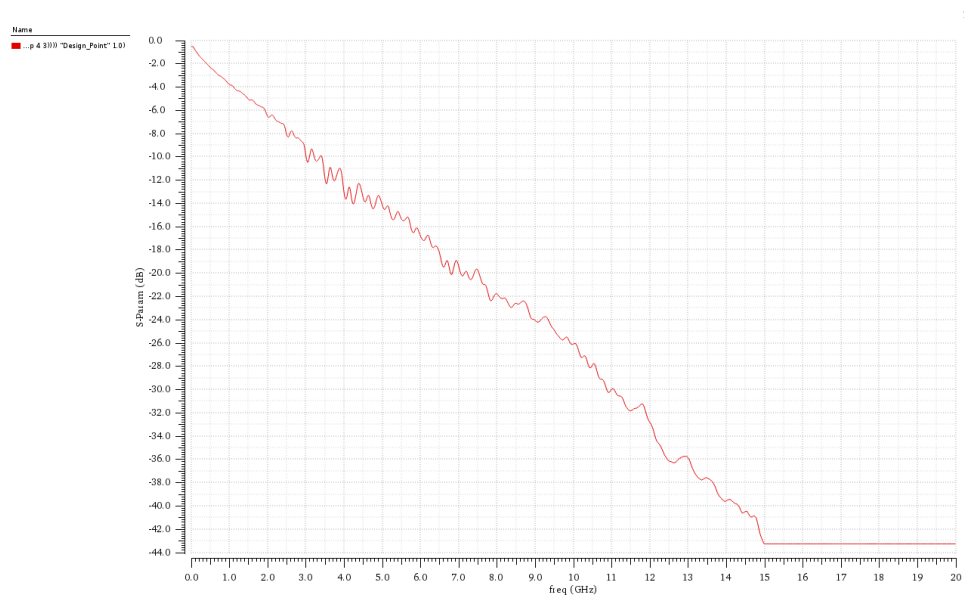


Figure 8: Method 2: Extracted SDD_{21} magnitude response.

3 Part 2: 8 Gb/s Transient Pulse Response

3.1 Simulation Methodology

Part 2 evaluates high-speed channel dispersion at 8 Gb/s (UI = 125 ps).

- **Setup:** 4-port channel with $50\ \Omega$ terminations (Figure 9).
- **Excitation:** Complementary pulses ($V_{\text{high}} = 0.5\ \text{V}$, $V_{\text{low}} = 0\ \text{V}$, pulse width = 125 ps, $t_r = t_f = 6.25\ \text{ps}$) starting at $t = 5.0\ \text{ns}$.
- **Simulation Parameters:** Transient sweep to 15 ns with conservative accuracy option and $\text{maxstep} = 1\ \text{ps}$.

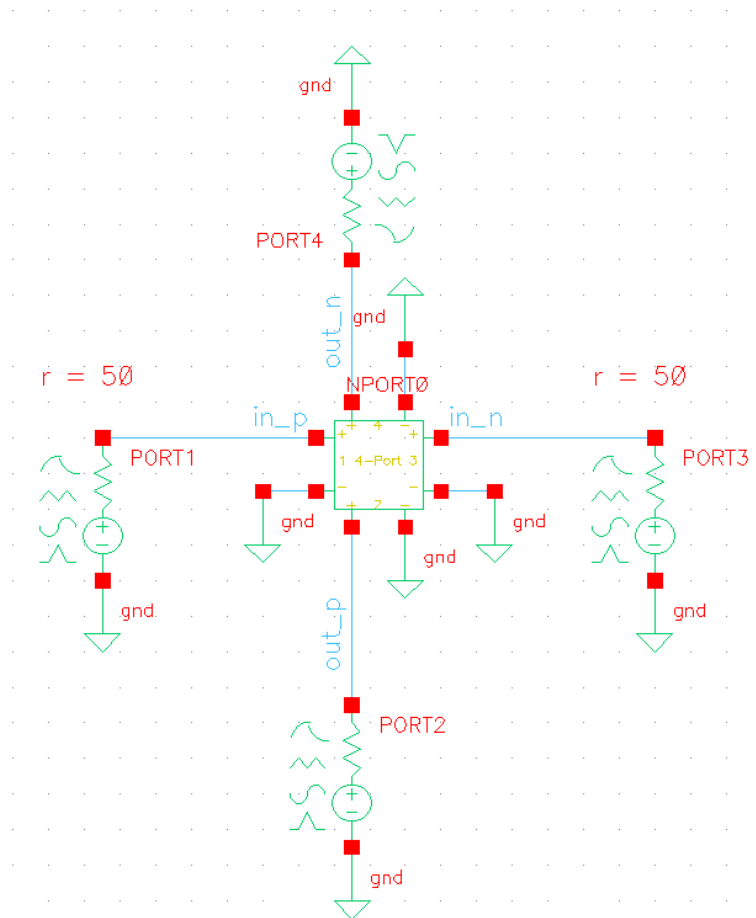


Figure 9: Transient pulse response simulation schematic.

3.2 Pulse Response Analysis and Cursor Extraction

The output signal $V_{\text{diff}}(t) = V(\text{out_p}) - V(\text{out_n})$ peaks at $t_{\text{main}} = 9.1043\ \text{ns}$ (522.8507 mV). Cursors are sampled at 1 UI = 125 ps intervals relative to t_{main} (Table 1).

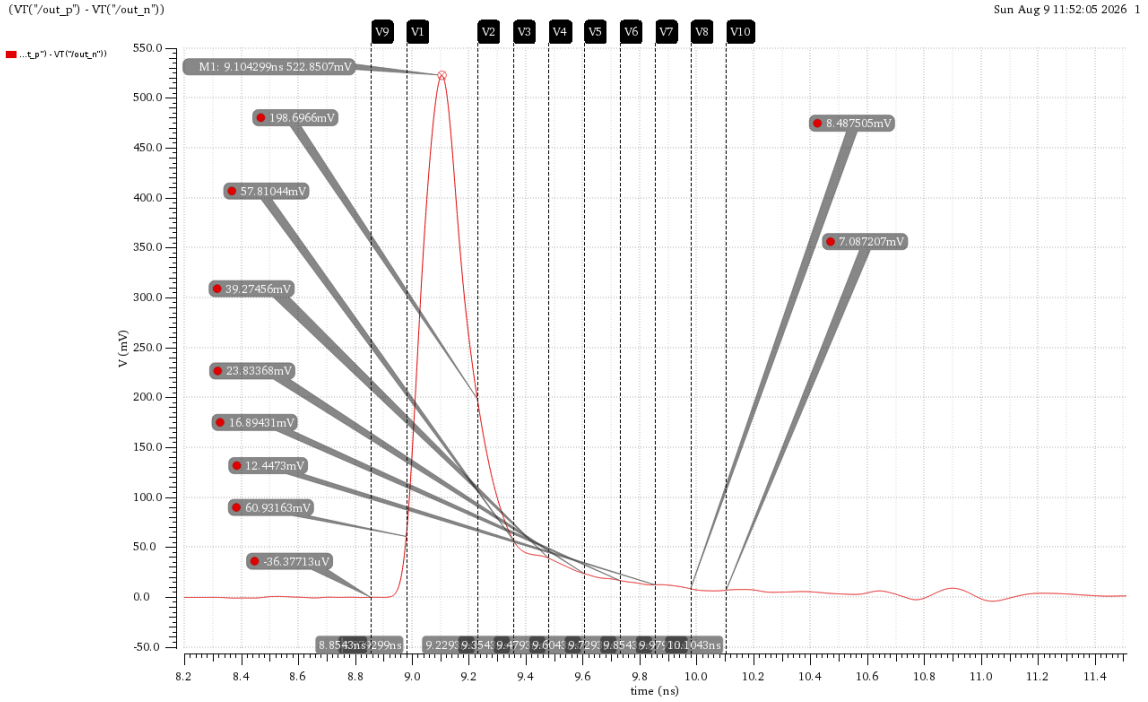


Figure 10: Differential transient pulse response $V(\text{out}_p) - V(\text{out}_n)$ with UI cursors.

Table 1: Extracted Pulse Response Cursors (1 UI = 125 ps)

Marker	Time (ns)	Cursor Type	UI Offset	Value
V9	8.8543	Pre-cursor 2	-2 UI	-36.37713 μV
V1	8.9793	Pre-cursor 1	-1 UI	60.93163 μV
M1	9.1043	Main Cursor	0 UI	522.8507 mV
V2	9.2293	Post-cursor 1	+1 UI	198.6966 mV
V3	9.3543	Post-cursor 2	+2 UI	57.81044 mV
V4	9.4793	Post-cursor 3	+3 UI	39.27456 mV
V5	9.6043	Post-cursor 4	+4 UI	23.83368 mV
V6	9.7293	Post-cursor 5	+5 UI	16.89431 mV
V7	9.8543	Post-cursor 6	+6 UI	12.44730 mV
V8	9.9793	Post-cursor 7	+7 UI	8.487505 mV
V10	10.1043	Post-cursor 8	+8 UI	7.087207 mV

3.3 Inter-Symbol Interference (ISI) Observations

- **Pre-cursors:** Pre-cursor distortion is negligible ($< 61 \mu\text{V}$), confirming causal channel characteristics prior to pulse arrival.
- **Post-cursors:** High post-cursor ISI is observed, with $h[+1] = 198.7 \text{ mV}$ (38% of peak height). Significant post-cursor residual voltage remains through $h[+8]$, highlighting the requirement for equalization (FFE/DFE).

3.3.1 Comments on the Pulse Response

Since a single pulse was applied at the channel input, the observed pre-cursor and post-cursor components are not caused by adjacent transmitted symbols. Instead, they represent the temporal spreading introduced by the channel itself. The main contributors to this spreading are the limited and frequency-dependent channel bandwidth, the dispersive phase response, and impedance discontinuities that produce reflections.

The frequency-dependent attenuation causes different spectral components of the pulse to be attenuated by different amounts, while the frequency-dependent phase response causes these components to experience different delays. Both effects contribute to pulse spreading around the main cursor. In addition, reflections caused by impedance mismatches can produce delayed replicas of the signal, which are observed as smaller post-cursor components.

4 Conclusion

This lab presented the full characterization of a high-speed multi-port channel. S-parameter analysis established the channel insertion loss and determined a group delay of $\tau_g \approx 4.00$ ns, closely tracking the physical propagation delay (~ 4.10 ns). Transient pulse simulations identified severe post-cursor ISI ($h[+1] = 198.7$ mV), yielding tap coefficients critical for system-level link optimization.